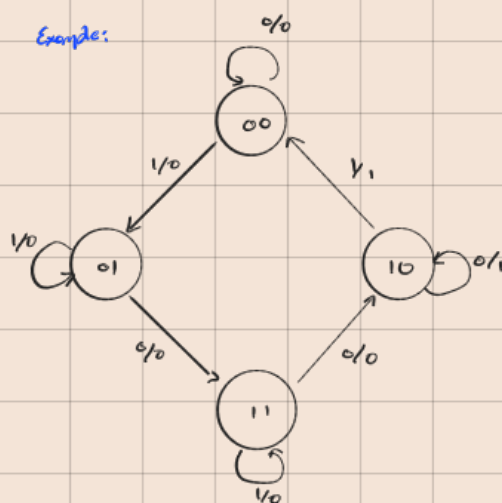


- 1) state diagram given / state table given
- 2) obtain state table / obtain state diagram
- 3) number of states reduced.
- 4) state Assignment if regular
- 5) number of D flip flops used
- 6) Type of flip flop used.
- 7) Circuit Equation table from state table
- 8) Expression for circuit output and ff input
- 9) Implement the circuit

Example:



State table

PS		NS		output	
Q _n	Q _{n-1}	n=0		n=1	
		Q _n	Q _{n-1}	Q _n	Q _{n-1}
0	0	0	0	0	0
0	1	1	1	0	0
1	0	1	0	1	1
1	1	1	0	0	0

∴ using T Flip Flops.

Excitation table

PS		NS	T		Q	
Q _n	Q _{n-1}	Q _{n+1}	T _n	T _{n-1}	Q _n	Q _{n-1}
0	0	0	0	0	0	0
0	0	1	0	1	0	1
0	1	0	1	1	1	0
0	1	1	0	1	1	1
1	0	0	1	0	0	0
1	0	1	0	0	1	0
1	1	0	1	0	0	1
1	1	1	0	1	1	1

Q _n	Q _{n-1}	T
0	0	0
0	1	1
1	0	1
1	1	0

$$T_A = \bar{Q}_n \bar{Q}_{n-1} + Q_n \bar{Q}_{n-1}$$

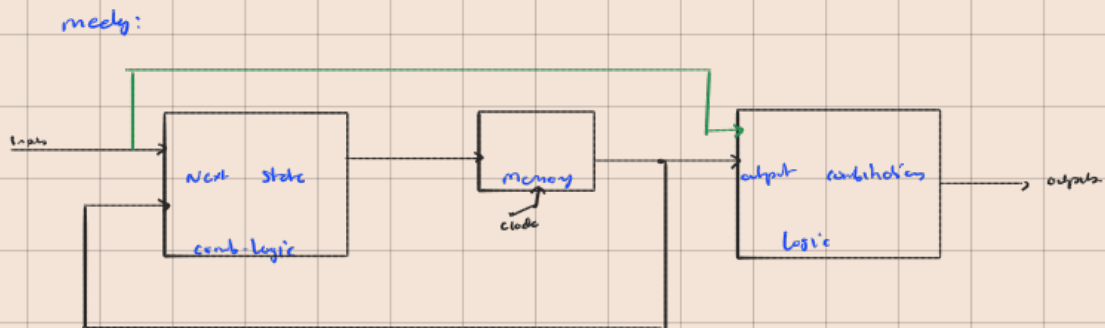
$$T_B = \bar{Q}_n \bar{Q}_{n-1} + Q_n Q_{n-1}$$

i) moore circuit / moore state machine :- o/p is the function of P.S only

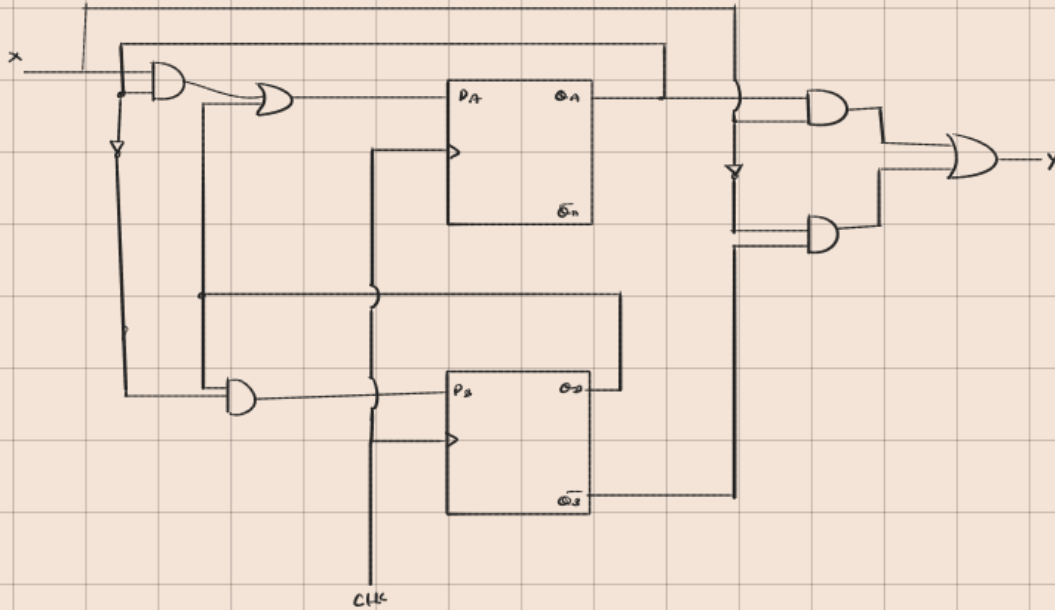
ii) mealy state machine :- o/p is the function of P.S as well as input



1a moore

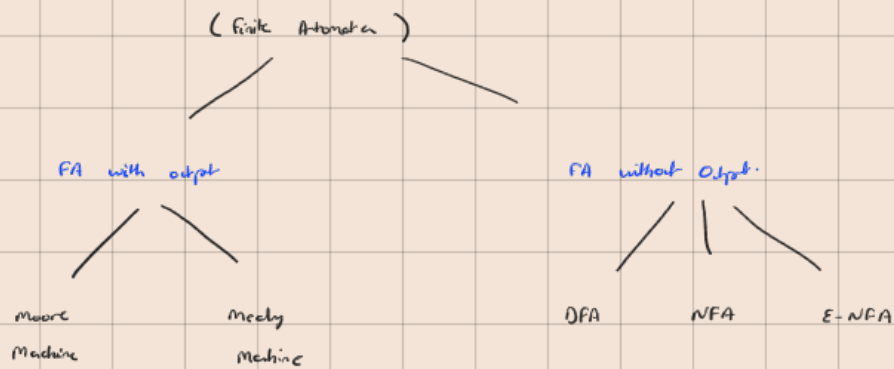


Analysis of (closed) Separable Circuits (with 10 flip flops)



step 1: find V_p & o/p equation

Finite state Machine:



Every DFA can be defined using

$(Q, \Sigma, q_0, F, \delta)$

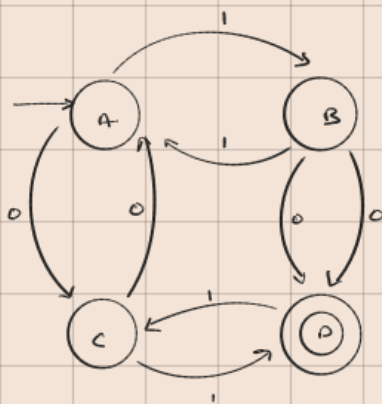
Q = set of all states

Σ = Inputs

q_0 = start state / initial state (only 1, represented by arrow coming from nowhere)

F = set of final states (can be more than 1)

δ = transition function from $Q \times \Sigma \rightarrow Q$



$Q = \{A, B, C, D\}$

$\Sigma = \{0, 1\}$

$q_0 = A$

$F = \{D\}$

δ	0	1
A	C	B
B	D	A
C	A	D
D	B	C

FSM (prerequisites)

Symbol: - $a, b, c, \dots, 1, 2, 3, \dots$

Alphabet: - Σ - collection of symbols Eg: $\{a, b\}$, $\{0, 1\}$, $\{a, b, c\}$

String: - sequence of symbols Eg: $a, b, a, 1, aa, bb, ab, 01,$

Language: - set of strings

Eg: $\Sigma = \{0, 1\}$

$L_1 =$ set of all string of length 2

$= \{00, 01, 10, 11\}$

Construction of Mealy Machine - Examples (Part-1)

Design a Mealy Machine accepting the language consisting of strings from Σ^* , where $\Sigma = \{a,b\}$ and the strings should end with either aa or bb

aa - 1
bb - 1

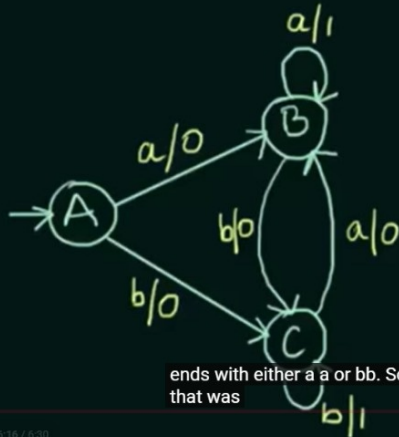


abb
baa

ba
00

aa
01

Construction of Mealy Machine - Examples (Part-2)

Construct a Mealy Machine that gives 2's Complement of any binary input. (Assume that the last carry bit is neglected)

2's complement = 1's complement + 1

Msb ← LSB

Eg. 10100

1's. 01011

+ 1

2's = 01100

Eg. 1100

1's. 0011

+ 1

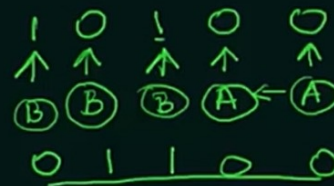
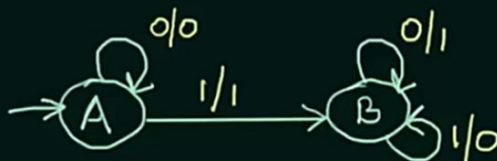
2's = 00100

Eg. 1111

1's. 0000

2's. + 1

0001

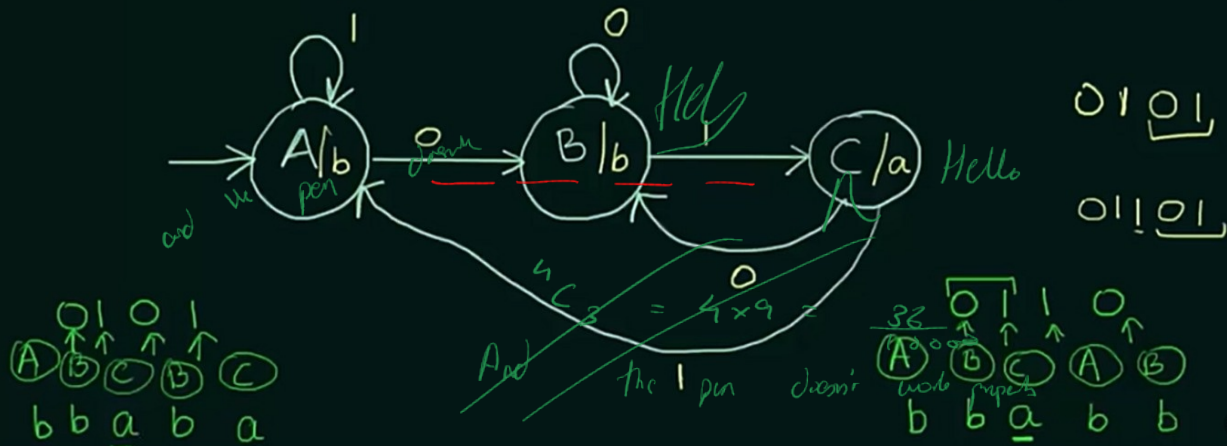


Construction of Moore Machine

Construct a Moore Machine that prints 'a' whenever the sequence '01' is encountered in any input binary string

$$\Sigma = \{0, 1\}$$

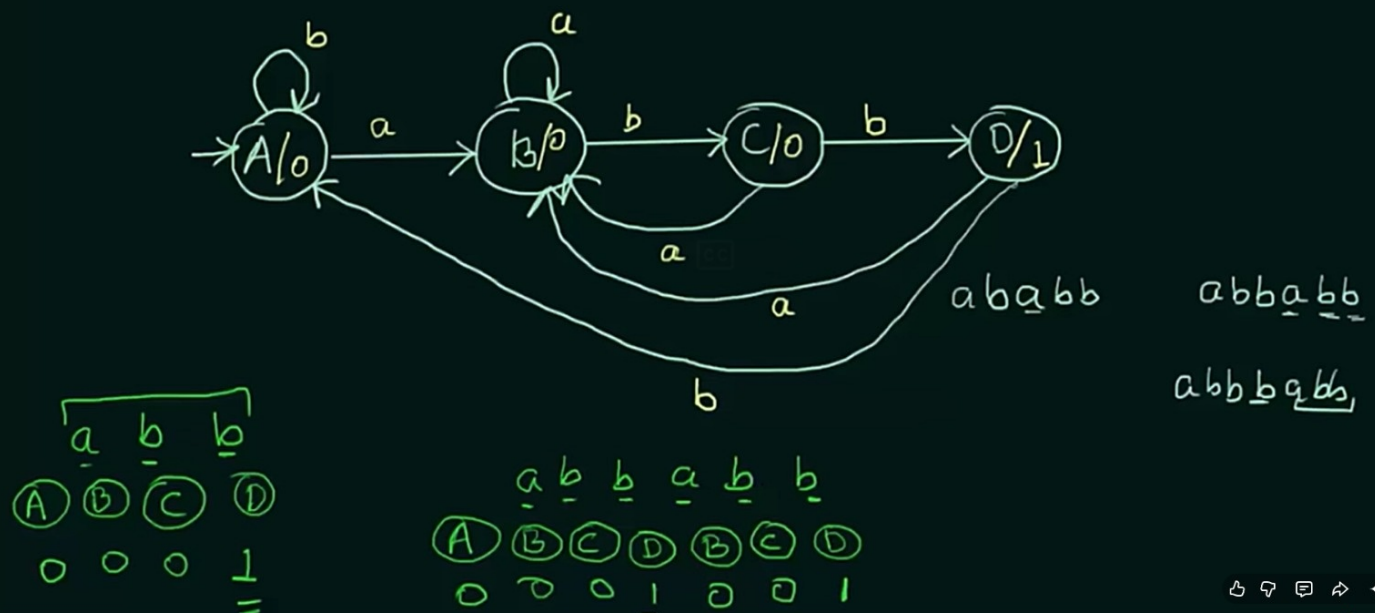
$$\Delta = \{a, b\}$$



Construct a Moore Machine that counts the occurrences of the sequence 'abb' in any input strings over {a,b}

$$\Sigma = \{a, b\}$$

$$\Delta = \{0, 1\}$$

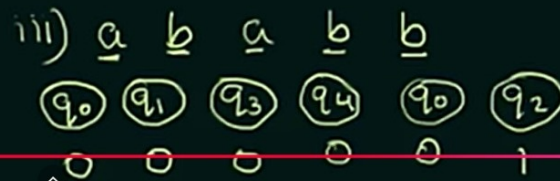
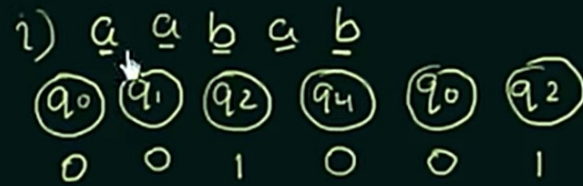


Beeba
♥
Arsha

For the following Moore Machine the input alphabet is $\Sigma = \{a, b\}$ and the output alphabet is $\Delta = \{0, 1\}$. Run the following input sequences and find the respective outputs:

(i) aabab (ii) abbb (iii) ababb

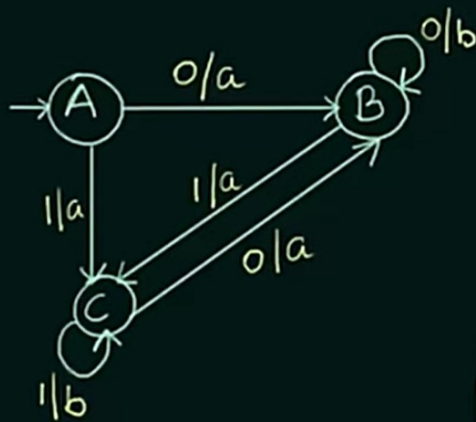
States	a	b	Outputs
$\rightarrow q_0$	q_1	q_2	0
q_1	q_2	q_3	0
q_2	q_3	q_4	1
q_3	q_4	q_4	0
q_4	q_0	q_0	0



Conversion of Mealy Machine to Moore Machine

Convert the following Mealy Machine to its equivalent Moore Machine

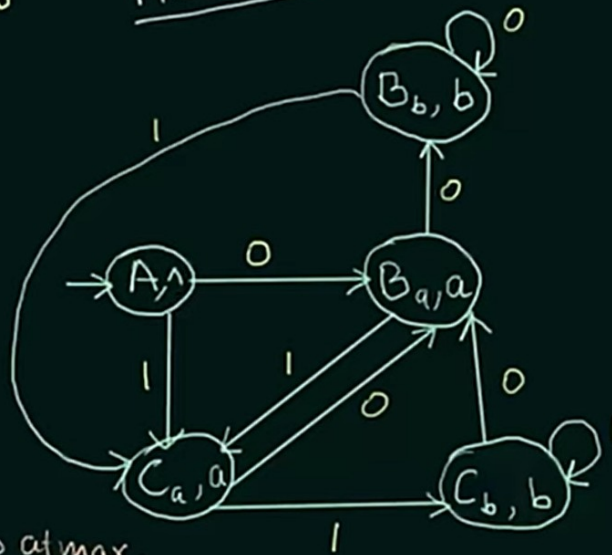
$\Sigma = \{0, 1\}$
 $\Delta = \{a, b\}$



Moore $\rightarrow$ Mealy $\Rightarrow$ No. of states were same
 Mealy $\rightarrow$ Moore $\Rightarrow$ No. of states increased

$\downarrow$
 n and $y \Rightarrow (n \times y)$ no. of states at max.
 $\downarrow$ $\downarrow$
 states outputs

Moore Machine



Conversion of Mealy Machine to Moore Machine -Examples (Part-3)

Using Transition Table

Convert the given Mealy Machine to its equivalent Moore Machine

State	a	b
$\rightarrow q_0$	$q_3, 0$	$q_1, 1$
q_1	$q_0, 1$	$q_3, 0$
q_2	$q_2, 1$	$q_2, 0$
q_3	$q_1, 0$	$q_0, 1$

Moore

State	a	b	Output
$\rightarrow q_0$	q_3	q_{11}	1
q_{10}	q_0	q_3	0
q_{11}	q_0	q_3	1
q_{20}	q_{21}	q_{20}	0
q_{21}	q_{21}	q_{20}	1
q_3	q_{10}	q_0	0

